

SYDNEY GIRLS HIGH SCHOOL
TRIAL HIGHER SCHOOL CERTIFICATE, 1993
3/4 UNIT MATHEMATICS

TIME ALLOWED: Two hours (including reading time)

DIRECTIONS TO CANDIDATES:

- * All questions may be attempted
- * All questions are of equal value
- * All necessary working should be shown in every question
Marks may not be awarded for careless or badly arranged work
- * Standard integrals are printed on a separate paper
- * Calculators may be used
- * Start a new page for each question

N.B. This is a TRIAL PAPER ONLY and does not necessarily reflect the content or format of the final Higher School Certificate Examination Paper for this subject

Question 1:

a) Solve $x - \frac{1}{x} < 0$

b) If $x = \sqrt{3} + \sqrt{2}$ find the value of $x + \frac{1}{x}$, hence or otherwise find the value of $\left(x^2 + \frac{1}{x^2}\right)$

c) The point P (6,9) divides the interval AB in the ratio -3:2. Find the point B given that A is (1,4)

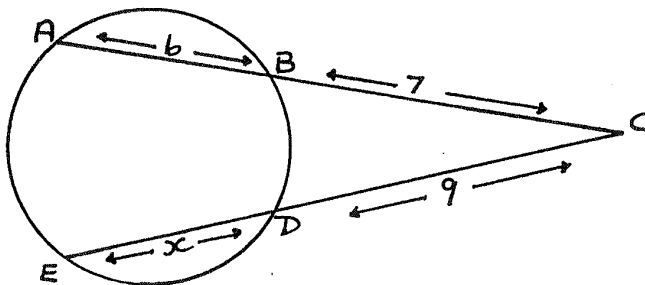
d) $\int \frac{-x}{\sqrt{1-x^2}} dx$

e) $\int \frac{3}{16+x^2} dx$

Question 2:

a) Solve $\sin x + \sin 2x = 0$ $0^\circ \leq x \leq 360^\circ$

b)



Find x, give reasons

c) The positive square root of 37 is approximately 6. Use one application of Newton's Method to find a better approximation, correct to 2 decimal places

d) A function is defined by the rules

$$f(x) = \begin{cases} \sin^{-1} x & \text{if } -1 \leq x < 0 \\ \cos^{-1} x & \text{if } 0 \leq x \leq 1 \end{cases}$$

i) Sketch the graph of the function for $-1 \leq x \leq 1$

ii) For this function evaluate exactly $f(\frac{1}{2}) + 2f(0) - f(-\frac{1}{2})$

Question 3:

- a) A certain particle moves along the x axis in accordance with the law

$$t = 2x^2 - 4x + 3$$

where x is measured in cms and t in seconds

Initially the particle is 1.5cms to the right of 0 and moving away from 0

- i) Prove that the velocity, v cm/sec is given by

$$v = \frac{1}{4x - 4}$$

- ii) Find an expression for the acceleration a cm/sec² in terms of x

- iii) Find the velocity and acceleration of the particle when

- a) x = 2cms
b) t = 9secs

- iv) Describe in words the motion of the particle

- b) Prove by induction that

$$3^n > 1 + 2n \text{ for } n > 1$$

Question 4:

The points P (2ap, ap²) and Q (2aq, aq²) lie on the parabola x² = 4ay

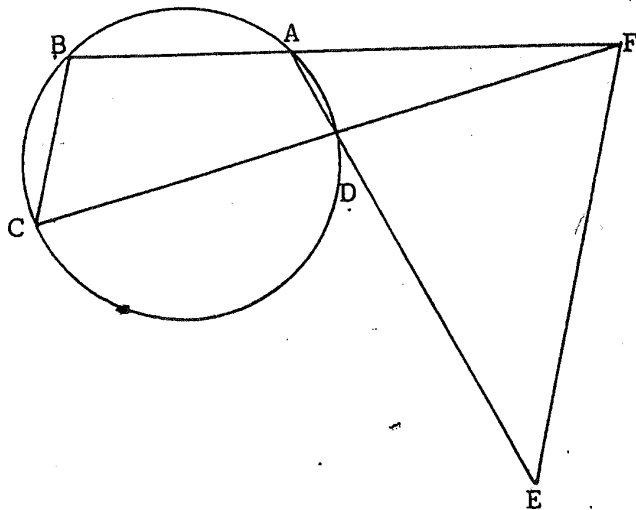
- a) Derive the equation of the normal to the parabola at P
b) Find the coordinates of the point N the intersection of the normals at P and Q
c) Find the equation of the chord PQ and determine the condition necessary for PQ to be a focal chord
d) If PQ is a focal chord and N is the intersection of the normals at P and Q, find the equation of the locus of N. Describe the locus

Question 5:

a) Prove that

$$\frac{1 - \cos \theta}{1 + \cos \theta} = \tan^2 \frac{\theta}{2}$$

b)



BC $\parallel$ EF

ABCD is a cyclic quadrilateral
Prove that

$$EF^2 = EA \times ED$$

c) The product of two of the roots of the equation

$$24x^3 + 14x^2 - 11x - 6 = 0$$

is equal to the other root. Find the values of all three roots of the equation

Question 6:

- a) Find the value of

$$\int_0^{\frac{\pi}{2}} \frac{\sec^2(\sin x) dx}{\sec x}$$

using the substitution

$$u = \sin x$$

- b) Express $7 \cos \alpha - \sin \alpha$ in the form $R \cos(\alpha + \beta)$ where $R > 0$ and $0^\circ < \beta < 360^\circ$. Hence solve the equation

$$7 \cos \alpha - \sin \alpha = 5$$

- c) Assume that the rate at which a body cools in air is proportional to the difference between its temperature T and the constant temperature A of the surrounding air. This rate can be expressed by the differential equation

$$\frac{dT}{dt} = -k(T - A)$$

where t is the time in minutes and k is a constant

- i) Show that $T = A + Ce^{-kt}$ is a solution to the differential equation (C is a constant)
- ii) The body of a murder victim is discovered at 2am when its temperature is 35°C . One hour later its temperature has fallen to 34°C . If the room temperature remains constant at 21°C , find the value of k
- iii) Calculate the time the murder was committed. (Normal body temperature is approximately 37°C .)

Question 7:

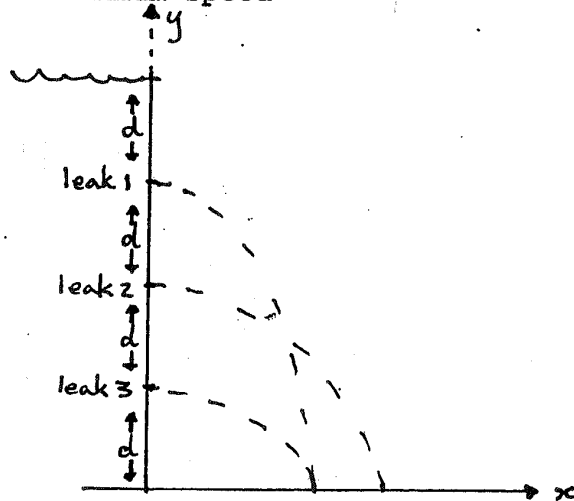
- a) Find the volume of the solid of revolution obtained by revolving the area bounded by $y = 1 + \sin x$ and the x -axis between $x = 0$ and $x = \pi$ about the x axis.

- b) A particle undergoes simple harmonic motion about the origin 0. Its displacement x cm from 0 at time t seconds is given by

$$x = 5 \cos \left(4t + \frac{\pi}{2} \right)$$

- i) Find the acceleration in terms of t and hence or otherwise as a function of displacement
- ii) Write down the amplitude of the motion
- iii) Find the maximum speed

c)



A dam wall is $4d$ units high. Small leaks occur at 3 evenly spaced points $3d$, $2d$ and $1d$ units down from the water level

- i) Using axes as shown, show that the parametric equations for the water from leak 1 are:

$$x = Vt$$

$$y = 3d - \frac{1}{2}gt^2$$

- ii) Find the parametric equations for leak 2 and leak 3
- iii) Show that the jets of water from leak 1 and leak 3 reach the ground level at the same point given that the velocity of water issuing from a leak is $V^2 = kh$ where h is the height of water above the leak and k is a constant
- iv) Find the velocity of impact of the water from leak 2 in terms of g , k and d
- v) A leak occurs at a distance l units above ground. Show that it hits the ground a maximum distance from the dam if $l = 2d$

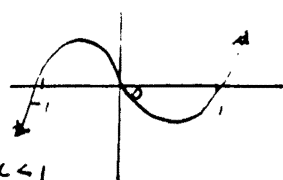
Hom 1

$$-\frac{1}{x} < 0$$

$$-x < 0$$

$$-1(x+1) < 0$$

$$-1 < 0 < x < 1$$



$$\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} = \frac{\sqrt{3}-\sqrt{2}}{1}$$

$$\frac{1}{\sqrt{3}+\sqrt{2}} = \frac{1}{\sqrt{3}+\sqrt{2}} \cdot \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} = \frac{\sqrt{3}-\sqrt{2}}{3-2} = \sqrt{3}-\sqrt{2}$$

$$\begin{aligned} +\frac{1}{x^2} &= \left(x + \frac{1}{x}\right)^2 - 2 \\ &= (2\sqrt{3})^2 - 2 \\ &= 12 - 2 \\ &= 10 \end{aligned}$$

$$AP:PB = -3:2$$

$$\frac{k_1 x_2 + k_2 x_1}{k_1 + k_2} = \frac{k_1 y_2 + k_2 y_1}{k_1 + k_2}$$

$$\begin{aligned} \frac{-3x_2 + 2x_1}{-1} &= \frac{-3y_2 + 2x_4}{-1} \\ &= 3x - 2 = 3y - 8 \\ &= \frac{8}{3} \end{aligned}$$

$$\therefore B\left(\frac{8}{3}, \frac{17}{3}\right)$$

$$\frac{-x}{\sqrt{1-x^2}} dx = \sqrt{1-x^2} + C$$

$$\begin{aligned} \frac{3 dx}{16+x^2} &= 3 \int \frac{dx}{16+x^2} \\ &= \frac{3}{4} \tan^{-1} \frac{x}{4} + C \end{aligned}$$

Section 2

$$\sin x + \sin 2x = 0$$

$$\sin x + 2\sin x \cos x = 0$$

$$\sin x (1 + 2\cos x) = 0$$

$$\sin x = 0 \quad \text{or} \quad 1 + 2\cos x = 0$$

$$x = 0^\circ, 180^\circ, 360^\circ \quad 2\cos x = -1$$

$$\cos x = -\frac{1}{2}$$

$$x = 120^\circ$$

$$x = 240^\circ$$

$$d) \text{ Let } f(x) = x^2 - 37 = 0$$

a root of which is exactly $\sqrt{37}$

Taking $x_1 = 6$ as a first approx.

$$\text{then } x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

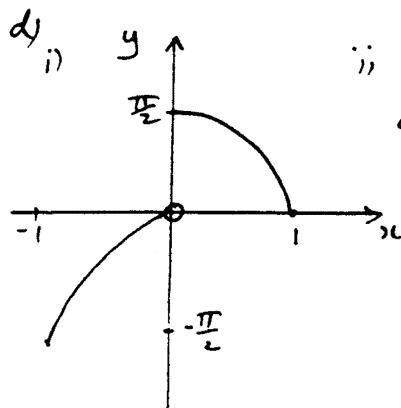
$$f'(x) = 2x$$

$$\therefore f(6) = -1 \text{ and } f'(6) = 12$$

$$x_2 = 6 - \frac{-1}{12}$$

$$= 6\frac{1}{12}$$

$$= 6.08 \text{ (2 decpl)}$$



$$ii) f\left(\frac{1}{2}\right) + 2f(0) - f\left(-\frac{1}{2}\right)$$

$$\cos^{-1} \frac{1}{2} + 2\cos^{-1} 0 - \sin^{-1} \frac{1}{2}$$

$$\begin{aligned} \frac{\pi}{3} + 2 \cdot \frac{\pi}{2} - \left(-\frac{\pi}{6}\right) \\ \frac{\pi}{3} + \pi + \frac{\pi}{6} \\ \frac{3\pi}{2} \end{aligned}$$

Question 3

$$i) t = 2x^2 - 4x + 3$$

$$\therefore \frac{dt}{dx} = 4x - 4$$

$$\text{Now } v = \frac{dx}{dt} = \frac{1}{\frac{dt}{dx}} = \frac{1}{4x-4}$$

$$\begin{aligned} ii) a &= v \frac{dv}{dx} \quad \left[a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} \right] \\ &= \frac{1}{4x-4} \cdot \frac{-1}{(4x-4)^2} \cdot 4 \\ &= \frac{-4}{(4x-4)^3} \end{aligned}$$

$$iii) \text{ when } x = 2$$

$$v = \frac{1}{4 \times 2 - 4} = \frac{1}{4} \text{ cm/sec}$$

$$a = \frac{-4}{(8-4)^3} = \frac{-4}{64} = -\frac{1}{16} \text{ cm/sec}^2$$

$$b) \text{ when } t = 9$$

$$2x^2 - 4x + 3 = 9$$

$$2x^2 - 4x - 6 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$y = a(p^2 + pq + q^2 + 2)$$

$$2q = -1 \quad x = a(p+q)$$

$$y = a(p^2 + q^2 + 1)$$

$$x^2 = a^2(p^2 + 2pq + q^2)$$

$$= a^2(p^2 + q^2 - 2)$$

$$q^2 = \frac{x^2}{a^2} + 2 = \frac{y}{a} = -1$$

Eq of locus of N is

$$x^2 = a(y - 3a)$$

is a parabola with
vertex $(0, 3a)$ and focus
 $(0, \frac{13}{4}a)$

Question 5

$$\frac{1 - \cos \theta}{1 + \cos \theta} = \tan^2 \frac{\theta}{2}$$

$$\frac{1 - (1 - 2\sin^2 \frac{\theta}{2})}{1 + (2\cos^2 \frac{\theta}{2} - 1)}$$

$$\frac{2\sin^2 \frac{\theta}{2}}{2\cos^2 \frac{\theta}{2}}$$

$$\tan^2 \frac{\theta}{2}$$

$$= A + S$$

$\widehat{BCD} = \widehat{EAF}$ (ext $\angle$ of cyclic quad equals interior opp)

$\widehat{EDC} = \widehat{DFE}$ (alternate $\angle$'s)
 $BC \parallel EF$

$$\widehat{FAF} = \widehat{DFE}$$

$\Delta AFG + DFE$

$$\widehat{FAF} = \widehat{DFE} \text{ (proved)}$$

$\widehat{EDC}$ is common

$\Delta AFE \parallel \Delta FDE$ (equiangular)

$$\frac{EF}{EA} = \frac{ED}{EF} \text{ (ratios of corresponding sides)}$$

$$EF^2 = EA \cdot ED$$

4)

If roots are α, β, α

then $\alpha\beta = 8$

$$\alpha + \beta + \alpha = \frac{-14}{24}$$

$$2\alpha + \alpha\beta + \beta\alpha = \frac{-11}{24}$$

$$2\alpha\beta = \frac{6}{24} = \frac{1}{4} \Rightarrow \alpha^2 = \frac{1}{4}$$

$$\therefore \alpha = \pm \frac{1}{2}$$

$$\text{testing } A(\frac{1}{2}) = -5 \quad A(-\frac{1}{2}) = 0$$

$\therefore 2x+1$ is a factor

$$A(x) = (2x+1)(12x^2 + x - 6)$$

$$= (2x+1)(3x-2)(4x+3)$$

$\therefore$ the roots are $-\frac{1}{2}, \frac{2}{3}, -\frac{3}{4}$

Question 6

$$a) \text{ Let } u = \sin x$$

$$x = 0 \quad u = 0$$

$$x = \frac{\pi}{2} \quad u = 1$$

$$\frac{du}{dx} = \cos x$$

$$dx = \frac{du}{\cos x}$$

$$= \sec x \, dx$$

$$\int_0^1 \frac{\sec^2 u \, du}{\sec u}$$

$$[\tan u]_0^1$$

$$\tan 1 - \tan 0$$

$$1.5574 \text{ (4 D.P.)}$$

$$b) 7 \cos \alpha - \sin \alpha = R \cos \alpha + R \sin \alpha$$

$$\therefore R \cos \alpha = 7 \quad \text{--- (1)}$$

$$R \sin \alpha = 1 \quad \text{--- (2)}$$

sq. (1) + (2) and add

$$R^2(\cos^2 \alpha + \sin^2 \alpha) = 50$$

$$R = \sqrt{50} \text{ (take the positive value)}$$

From (1) + (2)

$$\tan \alpha = \frac{1}{7} \quad \left(\begin{array}{l} \text{since } \cos \alpha + \sin \alpha \\ \text{the next line} \\ \text{is not used} \end{array} \right)$$

$$\alpha = 8.1^\circ$$

$$\therefore 7 \cos \alpha - \sin \alpha = \sqrt{50} \cos (12.4^\circ) = 6.2$$

$$i) T = 21 + Ce^{-kt}$$

$$\frac{dT}{dt} = -kCe^{-kt}$$

$$= -k(T-21)$$

$$A = 21^\circ \text{ \& when } t=0 \text{ } T=35^\circ$$

$$35 = 21 + Ce^{-k \cdot 0}$$

$$\therefore C = 14$$

$$\text{at } t=60 \text{ } T=34$$

$$34 = 21 + 14e^{-60k}$$

$$13 = 14e^{-60k}$$

$$\frac{13}{14} = e^{-60k}$$

$$\frac{13}{14} = -60k \ln e$$

$$\frac{\ln \frac{13}{14}}{-60} = k$$

$$A + Ce^{-kt}$$

$$= 21 + 14e^{-kt}$$

$$= 14e^{-kt}$$

$$= e^{-kt}$$

$$\frac{13}{14} = e^{-kt}$$

$$\therefore \div 108 \text{ mins}$$

under committed

48 mins before 2am

12.12 a.m.

Question 7

$$I = \pi \int y^2 dx$$

$$= \pi \int_0^\pi (1 + \sin x)^2 dx$$

$$= \pi \int_0^\pi (1 + 2\sin x + \sin^2 x) dx$$

$$= \pi \left[x - 2\cos x + \frac{x}{2} - \frac{\sin 2x}{2} \right]_0^\pi$$

$$= \pi \left[\pi + 2 - \pi - 0 - 0 - 2 + 0 - 0 \right]$$

ii)

$$i) \dot{x} = -20 \sin \left(4t + \frac{\pi}{2} \right)$$

$$\ddot{x} = -80 \cos \left(4t + \frac{\pi}{2} \right)$$

$$= -16 \left(5 \cos \left(4t + \frac{\pi}{2} \right) \right)$$

$$= -16 \times 5$$

$$ii) \text{ amp} = 5$$

$$iii) \text{ Max speed occurs at } x=0$$

$$\dot{x}^2 = n^2(a^2 - x^2)$$

$$= 16(25 - 0)$$

Max speed 20 cm/sec.

at max value $\sin \alpha = 1$

$$\dot{x} = -20 \times 1$$

$$c. i) \frac{d^2y}{dx^2} = -g$$

$$\frac{dy}{dx} = -gt + c$$

initial vertical motion $\frac{dy}{dx} = V \sin \alpha$ at $t=0$

$$\therefore V \sin \alpha = -gt + c$$

$$\text{i.e. } c = V \sin \alpha$$

$$\therefore \frac{dy}{dx} = -gt + V \sin \alpha$$

$$y = V \sin \alpha t - \frac{1}{2}gt^2 + k$$

$$\text{initially } y = 3d \text{ when } t=0 \text{ \& } \alpha=0$$

$$\therefore k = 3d$$

$$\therefore y = 3d - \frac{1}{2}gt^2$$

Horizontal motion

$$\frac{dx}{dt} = V \cos \alpha$$

$$x = V \cos \alpha t + k$$

$$\text{when } t=0 \text{ } x=0 \text{ } k=0$$

$$\therefore x = V \cos \alpha t$$

$$\text{initially } \alpha = 0$$

$$x = Vt$$

$$1) \text{ leak 2 } y = 2d - \frac{1}{2}gt^2$$

$$x = Vt$$

$$\text{leak 3 } y = d - \frac{1}{2}gt^2$$

$$x = Vt$$

$$) \text{ leak 1 } h = d$$

$$V^2 = kd$$

$$V = \sqrt{kd}$$

$$x = Vt$$

$$= \sqrt{kd}t$$

$$y = 3d - \frac{1}{2}gt^2$$

$$\text{when } y = 0 \quad gt^2 = 6d$$

$$t = \sqrt{\frac{6d}{g}}$$

$$\therefore x = \sqrt{kd} \cdot \sqrt{\frac{6d}{g}}$$

$$= \sqrt{\frac{6kd^2}{g}}$$

$$\text{leak 3 } h = 3d$$

$$V^2 = kh$$

$$V = \sqrt{3kd}$$

$$x = Vt$$

$$y = d - \frac{1}{2}gt^2$$

$$\text{when } y = 0$$

$$gt^2 = 2d$$

$$t^2 = \frac{2d}{g}$$

$$t = \sqrt{\frac{2d}{g}}$$

$$x = \sqrt{3kd} \times \sqrt{\frac{2d}{g}}$$

$$= \sqrt{\frac{6kd^2}{g}}$$

$\therefore$ Water from leak 1 & 3 reach the ground level at the same pt. i.e. where $x = \sqrt{\frac{6kd^2}{g}}$

$$\text{leak 2 } h = 2d$$

$$V^2 = 2kd$$

$$V = \sqrt{2kd}$$

$$y = 2d - \frac{1}{2}gt^2$$

$$\text{when } y = 0 \quad gt^2 = 4d$$

$$t = \sqrt{\frac{4d}{g}}$$

$$\text{water reaches ground when } t = \sqrt{\frac{4d}{g}}$$

$$V^2 = \dot{x}^2 + \dot{y}^2 \quad \dot{y} = -gt$$

$$= -g\sqrt{\frac{4d}{g}} = -\sqrt{4dg}$$

$$\dot{x} = V = \sqrt{2kd}$$

$$\therefore V^2 = 2kd + 4dg$$

v) Max value of x

$$x = Vt \quad y = l - \frac{1}{2}gt^2$$

$$\text{water hits ground when } t = \sqrt{\frac{2l}{g}}$$

$$\therefore x = \sqrt{k(4d-l)} \times \sqrt{\frac{2l}{g}}$$

$$= \sqrt{\frac{8lkd - 2kd^2}{g}}$$

x is max when $\dot{x} = 0$ & when x^2 is m.

$$x^2 = \frac{8lkd - 2kd^2}{g}$$

$$\frac{dx^2}{dl} = \frac{8kd - 2kd}{g}$$

$$\frac{dx^2}{dl} = 0 \text{ when } 8kd = 4kd$$

$$\text{i.e. } 2d = l$$

$$\frac{d^2x^2}{dl^2} = -\frac{4k}{g} -ve$$